

Labs





About the Course

This is the required lab component for level 4 General Science. This course includes our custom laboratory book which will build foundational and progressive skills and habits.

This topic is included in the following course(s): Science: Grade 4

About Science: Grade 4

In level 4, learners begin to notice science in their community and begin to develop their own interests as they discover a variety of topics from the microscopic world, physical science, and more. A mixture of more and less challenging books is included in this course.

Science: Grade 4

The Big Picture:

To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Building on the familiarity and foundational knowledge of Form 1, Form 2 learners in grades 4-6 extend the scope of the Things they encounter and begin to notice that science is active in their world. Growing in their self-knowledge at this age, they may notice that they gravitate toward some of these Things more than others. This is a bit like noticing that you have a natural rapport with a particular friend because they seem to understand the way you think and share the same interests. This is wonderful and very typical, but we still want them to get to know more “friends” in science and try different Things, even though one might be their particular interest. Their sense of community is expanding, as well: who are the scientists in their world, and what are they doing? Special care has been taken to build vocabulary and conceptual knowledge “by the way” and to maintain the connection between scientific endeavors and the human experience. In Form 3, students will continue to build on this knowledge of what science is doing in the world when they examine science in its historical context.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just “feels right.” Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).



Placement & Combining Tips

About Science: Grade 4

In level 4, learners begin to notice science in their community and begin to develop their own interests as they discover a variety of topics from the microscopic world, physical science, and more. A mixture of more and less challenging books is included in this course.

Labs: Grade 4

The ideas and skills in this component are progressive, like math or grammar. Teachers should read the lab book thoroughly to understand what concepts might need to be supplemented should they choose a different sequence or a substitution.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
4	Labs: Grade 4 1 time/week 30 min	Science: Grade 4 Lab Book

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 4				
Natural History: Grade 4		General Science: Grade 4 Nature Walks & Scouting: Grades 1-8		Labs: Grade 4 Nature Notebook: Grade 4



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 4

- ☐ Obtain any supplies indicated on the science or grade-level supply lists.
- ☐ Download any apps and shortcut any desired links.

Nature Lore:

- ☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
- ☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Labs: Grade 4

- ☐ Preview science labs and purchase materials, or have students gather them. Print or shortcut any links, as desired.

Special Topics & Field Trips

Science: Grade 4

Encourage students to begin choosing their own special topic to observe, study, and research in their field guides. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too! Suggested field trips include:

Term Prep & Teacher Tips

Science: Grade 4

Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term and/or any that support special topics chosen by students:



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Labs: Grade 4



Science: Grade 4 Lab Book



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

Labs: Grade 4



Hand Lens



Small Collection Containers



Slides and Coverslips



Safety Goggles



Quick Links

Labs: Grade 4

∞ [Grade 4 Lab Book](#)

∞ [Lab Notebook Examples](#)

Click THIS text
or scan the QR
code for links.



Related Course Quick Links

∞ [Outdoor Work](#)

∞ [Microscope Coloring Book](#)

∞ [Seek app from iNaturalist](#)

∞ [SkyView Lite for iOS](#)

∞ [SkyView Lite for Android](#)

∞ [Foundations \(See Section 13: Science\)](#)

Labs: Grade 4

How To Teach Labs



Introduce

Regardless of how many days are required to complete a particular activity, every science lab has the same flow, which follows the scientific method and is guided by the lab book.

- On day 1, learners read an introduction in their lab book. How does this lab activity relate to what they have learned so far, as well as any previous experience? What will they learn from the lab? This is analogous to the conversation we might have when we begin something new in any subject, but they may need to dialogue as they learn to extend these skills to the laboratory.

- Once they have had a chance to think, they will compose a prelab narration to put these introductory ideas into their notebooks. The prompt in the lab is generalized and consistent, so they learn the habit. Eventually, they will learn to formulate this as a hypothesis. For example, a learner preparing for a lab about the use of insect repellent might write:

“I have read about some diseases that are spread by insects, like Lyme disease. I also know that my sister is allergic to some insect bites. Insect repellent contains pesticides to keep insects away. Some scientists worry about how pesticides affect wildlife. I am going to compare some different insect repellants in this lab to see if they really work.”

- Written narration and composition are skills that they will build over time. These prelab narrations may seem short and even incomplete at first, but that is okay. If learners have difficulty or are easily frustrated, then provide them with support. Teachers may act as scribes or allow students to keep a digital notebook to type or use assistive technology, as appropriate.

- After they complete their prelab narration, the listed materials are collected. This gives the learner some responsibility to let the teacher know if something is missing or to remind the teacher if something needs to be purchased at the store.

- If these activities on day 1 do not fill the scheduled time, that is fine. They might use the additional time to familiarize themselves with the procedure, draw a picture from their book, or catch up on any other work. Some labs may instruct the student to begin on the same day.



Lab Procedure

- Then, students begin and follow the procedure (whenever prompted by the lab book). Note that the lesson plans guide teachers as to which sections are completed each week, and the lab book instructs learners when to take a break.

- The lab gives instructions for using their notebooks to create tables and figures, as needed. Do not allow this to become an obstacle. Do it with them, first having them watch and then having them copy or help when ready.

- If teachers choose to have learners record directly in the lab book or on a photocopy, then cut out and tape these into their lab books, so that they can see how the record is built.

- Learners may feel unsatisfied with their results. This is often part of the process. Come alongside, help them learn to be okay with uncertainty and questions, and teach them what to do with it in the next step.



Analysis & Conclusions

- The last step in the lab is to analyze the data and observations and draw conclusions from them. For some simpler labs, learners will complete their analysis and concluding (or postlab) narration on the last day of the lab procedure. For labs that are more involved, a separate day has been built in to allow adequate time for this.

- Similar to the prelab narration, the concluding narration is a chance to think about and put into

words (or questions). This time, they are considering what they learned from the lab, what they could learn more about if they were to continue, and possibly how they would pursue that learning. Again, they might need support in the form of dialogue, a scribe, etc. For example, the above learner might write:

“It was clear that the Off and black pepper essential oil worked against ants because they would not even touch the line of repellent, but I wasn’t sure about the Skin So Soft because it ran all over the place. I could test this one again with different insects or on a different surface.”

- Depending on the interest of the learner and the priorities of the teacher, the student might be encouraged to spend more time on those ideas of what more they could learn, or it might be time to move on. Either way, it is an important part of the scientific method to reflect on what we could or would do next - our practice should help to clarify our thinking and teach that there is always more to be learned.
- Teachers should engage learners with this reflection by reviewing their lab notebooks with them, discussing the science used in the lab, and demonstrating curiosity about the lab themselves.

Labs: Grade 4

[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 1 ☐ 30m Labs: Grade 4 - Lesson 1

Lab: Meeting the Microscope

☐ Materials: Grade 4 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of Meeting the Microscope, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in the lab notebook. These need not be more than 1-3 sentences at first, and teachers should feel free to scribe for students, as necessary. Spend remaining time gathering materials for next week.

★ TEACHER TIP

Pacing is only a suggestion. Students should engage with the lab at a pace that is appropriate for their abilities and interests.

WEEK 2 ☐ 30m Labs: Grade 4 - Lesson 2

Lab: Meeting the Microscope

☐ Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of Meeting the Microscope, as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today, drawing, diagramming, or dictating in their lab notebook what they observe, just as they would in their nature journals.

WEEK 3 ☐ 30m Labs: Grade 4 - Lesson 3

Lab: Meeting the Microscope

☐ Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 3 of Meeting the Microscope, as directed in the Lab Book.

Suggested day 3: If students have a few last observations to make before completing their notebook, give them a few minutes to do so. Complete the postlab narration, as directed in the Analysis and Conclusion section of the lab. Teachers may scribe, journal in tandem, provide digital tools, or provide any other support as needed. If time remains, students may add a diagram of their microscope.

WEEK 4 ☐ 30m Labs: Grade 4 - Lesson 4

Lab: What's in Your Water?

☐ Materials: Grade 4 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of What's in Your Water?, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. Spend the remaining time gathering materials for next week.

★ TEACHER TIP

If needed, be sure to pour your agar plates for next week according to the package instructions.

∞ **Video Link:** Pouring Agar Plates (2:51)



Term 1

WEEK 5 30m Labs: Grade 4 - Lesson 5

Lab: *What's in Your Water?*

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of *What's in Your Water?*, as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today, drawing, diagramming, or dictating in their lab notebook what they observe, just as they would in their nature journals.

WEEK 6 30m Labs: Grade 4 - Lesson 6

Lab: *What's in Your Water?*

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 3 of *What's in Your Water?*, as directed in the Lab Book.

Suggested day 3: Students complete the Analysis and Conclusion instruction in the lab book, recording in their lab notebook just as they would in their nature journals. Teachers may scribe, journal in tandem, provide digital tools, or provide any other support as needed.

WEEK 7 30m Labs: Grade 4 - Lesson 7

Lab: *What's in Your Mouth?*

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 1 of *What's in Your Mouth?*, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. Spend the remaining time gathering materials for next week.

WEEK 8 30m Labs: Grade 4 - Lesson 8

Lab: *What's in Your Mouth?*

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of *What's in Your Mouth?*, as directed in the Lab Book.
∞ Video Link: *How to Dry Mount Cheek Cells* (4:07)

Suggested day 2: Students conduct the Procedure today. They will need help fixing and staining their slides. Then they may draw, diagram, or dictate in their lab notebook what they observe, just as they would in their nature journals. Today they will continue to the end, completing the short Analysis and Conclusion with a brief narration.

WEEK 9 30m Labs: Grade 4 - Lesson 9

Lab: *Explore and Discover!*

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 1 of *Explore and Discover!*, as directed in the Lab Book.

★ TEACHER TIP

If needed, be sure to pour your agar plates for next week according to the package instructions.

∞ Video Link: *Pouring Agar Plates* (2:51)

Labs: Grade 4

[Click THIS text or scan the QR code for links.](#)



Term 1

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. Students may collect their materials and begin the Procedure, as time permits.

WEEK 10 30m Labs: Grade 4 - Lesson 10

Lab: Explore and Discover!

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of Explore and Discover!, as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today, drawing, diagramming, or dictating in their lab notebook what they observe, just as they would in their nature journals.

WEEK 11 30m Labs: Grade 4 - Lesson 11

Lab: Explore and Discover!

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 3 of Explore and Discover!, or catch up on any other lab.

Suggested day 3: Students complete the Analysis and Conclusion instruction in the lab book, finishing any remaining observations and completing the postlab narration. Teachers provide whatever support is needed.

WEEK 12 30m Labs: Grade 4 - Lesson 12

Lab Exam Week

→ EXAMS

Answer question(s) related to the course.

Questions will come from:
Grade 4 Lab Book



Term 2

WEEK 1 30m Labs: Grade 4 - Lesson 13

Lab: Simple Machine Scavenger Hunt

 Materials: Grade 4 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of Simple Machine Scavenger Hunt, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. These need not be more than 1-3 sentences at first, and teachers should feel free to scribe for students, as necessary. Any remaining time can be spent copying the drawings from the Lab Book into their notebook.

★ TEACHER TIP

Pacing is only a suggestion. Students should engage with the lab at a pace that is appropriate for their abilities and interests.

WEEK 2 30m Labs: Grade 4 - Lesson 14

Lab: Simple Machine Scavenger Hunt

 Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of Simple Machine Scavenger Hunt, as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today. They may need help learning how to use the grid lines in their notebook to create a table for data collection. Students not ready to do so may use the table in the Lab Book and then cut and paste into their notebook.

WEEK 3 30m Labs: Grade 4 - Lesson 15

Lab: Simple Machine Scavenger Hunt

 Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 3 of Simple Machine Scavenger Hunt, as directed in the Lab Book.

Suggested day 3: Students finish the lab today using the Analysis and Conclusions section. They may need help learning how to construct a bar graph using the grid lines in their notebook, if they have not done this before. Instructions are provided in the Lab Book. If needed, students may use the table in the Lab Book and then cut and paste into their notebook. Then complete the postlab narration, as directed.

WEEK 4 30m Labs: Grade 4 - Lesson 16

Lab: What's an Aerofoil

 Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 1 of What's an Aerofoil, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. Any remaining time can be spent collecting the materials and copying the drawing from the Lab Book into their notebook.



Term 2

WEEK 5 ☐ 30m Labs: Grade 4 - Lesson 17

Lab: What's an Aerofoil

☐ Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of What's an Aerofoil, as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today, drawing, diagramming, or dictating in their lab notebook what they observe, just as they would in their nature journals.

WEEK 6 ☐ 30m Labs: Grade 4 - Lesson 18

Lab: What's an Aerofoil

☐ Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 3 of What's an Aerofoil, as directed in the Lab Book.

Suggested day 3: Students finish the lab today, as guided in the Analysis and Conclusions section.

WEEK 7 ☐ 30m Labs: Grade 4 - Lesson 19

Lab: Paper Airplanes

☐ Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 1 of Paper Airplanes, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. Any remaining time can be spent collecting the materials and copying the drawing from the Lab Book into their notebook.

WEEK 8 ☐ 30m Labs: Grade 4 - Lesson 20

Lab: Paper Airplanes

☐ Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of Paper Airplanes, as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today. They may need help learning how to use the grid lines in their notebook to create a table for data collection. If needed, students may use the table in the Lab Book and then cut and paste into their notebook.

WEEK 9 ☐ 30m Labs: Grade 4 - Lesson 21

Lab: Paper Airplanes

☐ Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 3 of Paper Airplanes, as directed in the Lab Book.

Suggested day 3: Many students want to collect another set of flight data or compare it with another airplane design. If so, let them do so and test



Term 2

their ideas. Challenge them to think about what is causing any differences that they see in their data. Students who are ready to move on may skip ahead to next week.	
WEEK 10 30m Labs: Grade 4 - Lesson 22 Lab: Paper Airplanes Materials: Grade 4 Lab Book and materials listed within → LAB DAY Complete day 4 of Paper Airplanes, as directed in the Lab Book. Suggested day 4: Students finish the lab today using the Analysis and Conclusions section. They may need help learning how to construct a bar graph using the grid lines in their notebook. Instructions are provided in the Lab Book. If needed, students may use the table in the Lab Book and then cut and paste into their notebook. Then complete the postlab narration, as directed.	
WEEK 11 30m Labs: Grade 4 - Lesson 23 Lab: Wings for Flight Materials: Grade 4 Lab Book and materials listed within → LAB DAY Complete the Wings for Flight activity, as directed in the Lab Book, or use this day to catch up on any other work.	
WEEK 12 30m Labs: Grade 4 - Lesson 24 Lab Exam Week → EXAMS Answer question(s) related to the course. Questions will come from: Grade 4 Lab Book	



Term 3

WEEK 1 30m Labs: Grade 4 - Lesson 25

Lab: Seeing Sound

 Materials: Grade 4 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of Seeing Sound, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. These need not be more than 1-3 sentences at first, and teachers should feel free to scribe for students, as necessary. Any remaining time can be spent gathering materials and/or copying the drawings from the Lab Book into their notebook.

★ TEACHER TIP

Pacing is only a suggestion. Students should engage with the lab at a pace that is appropriate for their abilities and interests.

WEEK 2 30m Labs: Grade 4 - Lesson 26

Lab: Seeing Sound

 Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of Seeing Sound, as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today. They may draw, diagram, or dictate in their lab notebook what they observe, just as they would in their nature journals. Then they will continue to the end, completing the short Analysis and Conclusion with a brief narration.

WEEK 3 30m Labs: Grade 4 - Lesson 27

Lab: Sending Sound

 Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 1 of Sending Sound, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. Any remaining time can be spent collecting the materials and possibly beginning the Procedure up to step 3.

WEEK 4 30m Labs: Grade 4 - Lesson 28

Lab: Sending Sound

 Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of Sending Sound, as directed in the Lab Book.

Suggested day 2: Students complete the Procedure today as directed in the Lab Book. Then they will continue to the end, completing the short Analysis and Conclusion with a brief narration.

WEEK 5 30m Labs: Grade 4 - Lesson 29

Lab: The Sound of Your Voice

 Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 1 of The Sound of Your Voice, as directed in the Lab Book.



Term 3

Suggested day 1: Students read the Introduction, compose the prelab narration, and complete step 1 of the Procedure, as directed in the Lab Book.

WEEK 6 30m Labs: Grade 4 - Lesson 30

Lab: The Sound of Your Voice

 Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of The Sound of Your Voice, as directed in the Lab Book.

Suggested day 2: Students complete the Procedure today, using the grid lines in their notebook to create a table for data collection. Offer support as needed.

WEEK 7 30m Labs: Grade 4 - Lesson 31

Lab: The Sound of Your Voice

 Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 3 of The Sound of Your Voice, as directed in the Lab Book.

Suggested day 3: Students finish the lab today using the Analysis and Conclusions section. Offer support as needed to construct a bar graph using the grid lines in their notebook. Then complete the postlab narration, as directed.

WEEK 8 30m Labs: Grade 4 - Lesson 32

Lab: Talking Animals

 Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 1 of Talking Animals, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and compose the prelab narration. Then they review the Procedure and decide when to begin based on the animal's needs.

WEEK 9 30m Labs: Grade 4 - Lesson 33

Lab: Talking Animals

 Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of Talking Animals, as directed in the Lab Book.

Suggested day 2: Students continue the Procedure today, as directed in the Lab Book.

WEEK 10 30m Labs: Grade 4 - Lesson 34

Lab: Talking Animals

 Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY

Complete day 3 of Talking Animals, as directed in the Lab Book.

Suggested day 3: Students who need to complete the Procedure today may do so and then finish with the Analysis and Conclusions section.

Term 3

WEEK 11 30m Labs: Grade 4 - Lesson 35 Lab: <i>Catch Up</i> Materials: Grade 4 Lab Book and materials listed within → CATCH UP Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	
WEEK 12 30m Labs: Grade 4 - Lesson 36 Lab <i>Exam Week</i> → EXAMS Answer question(s) related to the course. Questions will come from: Grade 4 Lab Book	